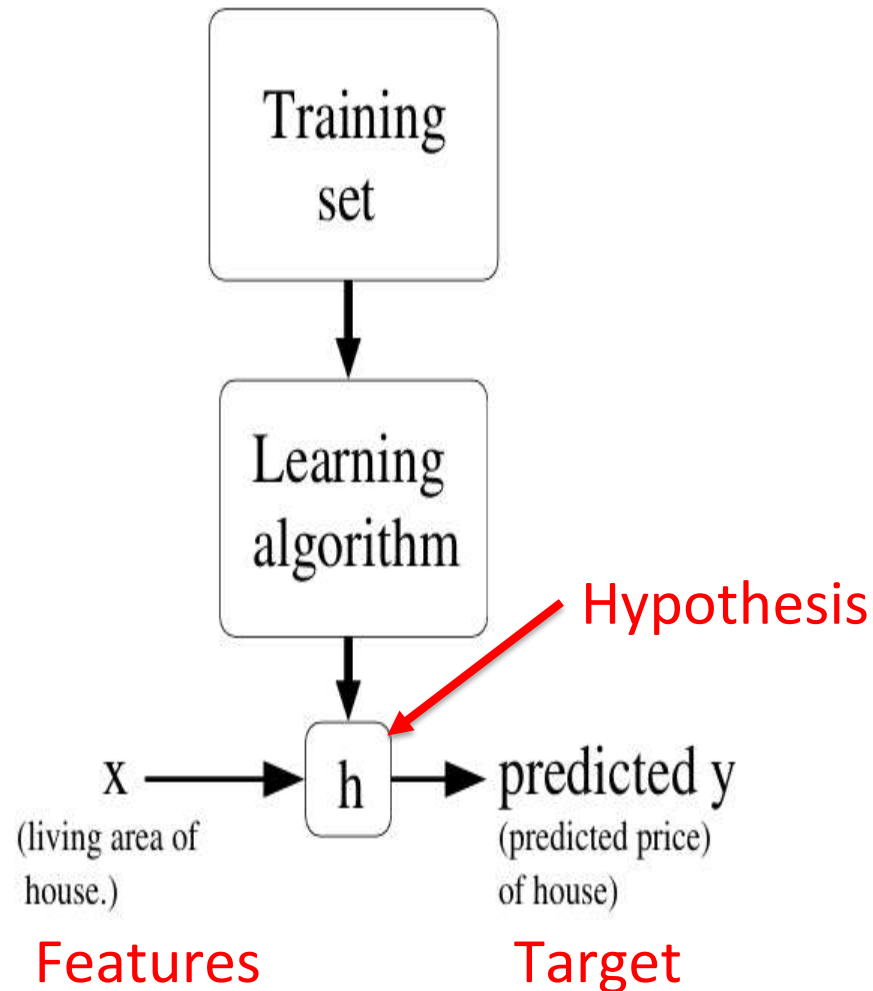


Linear Regression

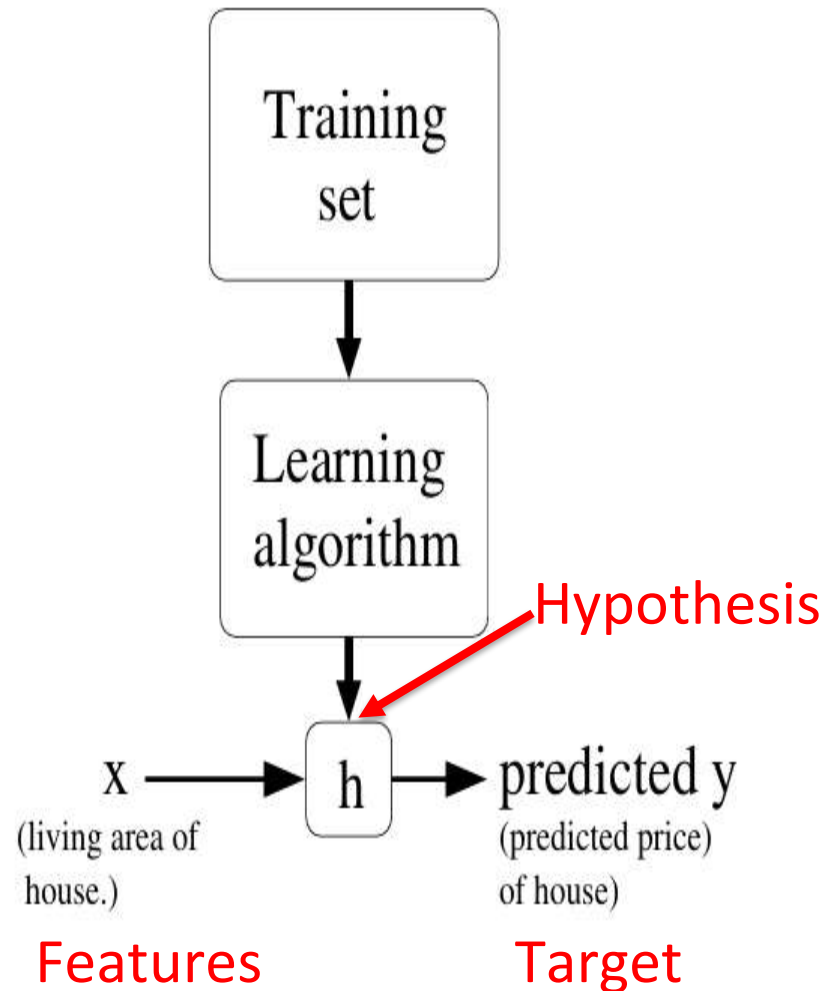
CSE 329

Supervised Learning (Recap)



Supervised Learning

- $x^{(i)}$: input or feature
- $y^{(i)}$: output or target
- $(x^{(i)}, y^{(i)})$: i^{th} training example
- $\{(x^{(i)}, y^{(i)}): i = 1, 2, \dots, n\}$ the training set
- Goal??



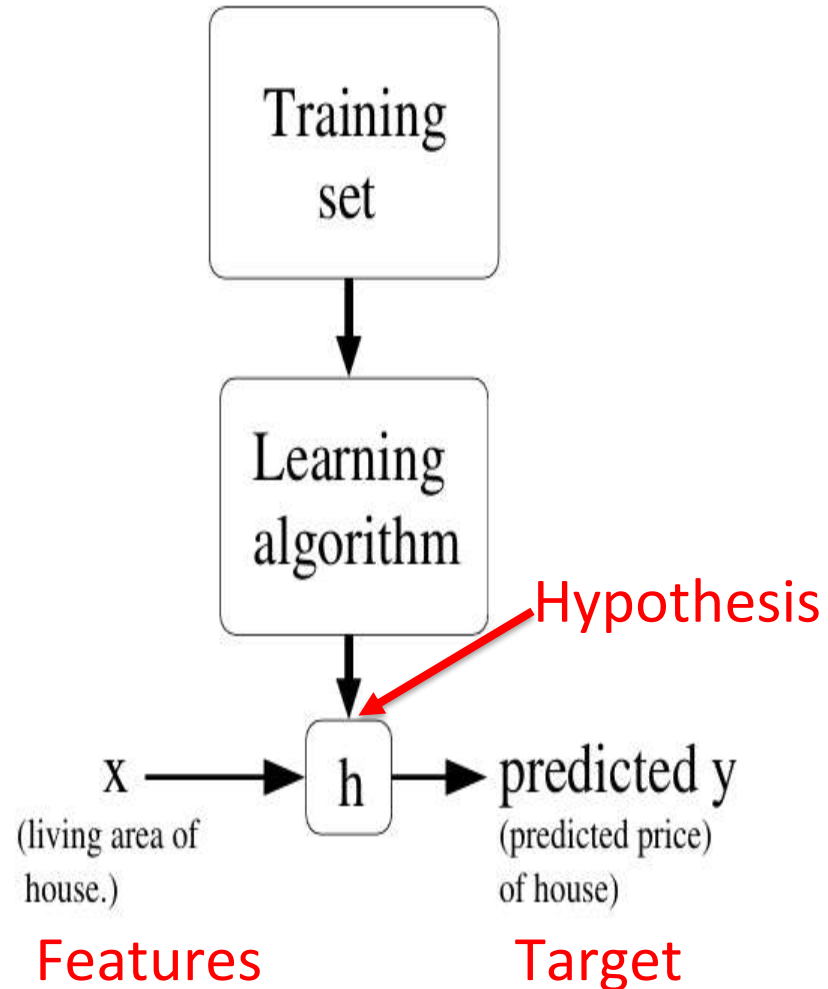
Supervised Learning

- $\mathbf{X}$ is the input space
- $\mathbf{y}$ is the target space
- Learn a **function** or **hypothesis** such that

$$h: X \rightarrow Y$$

i.e.

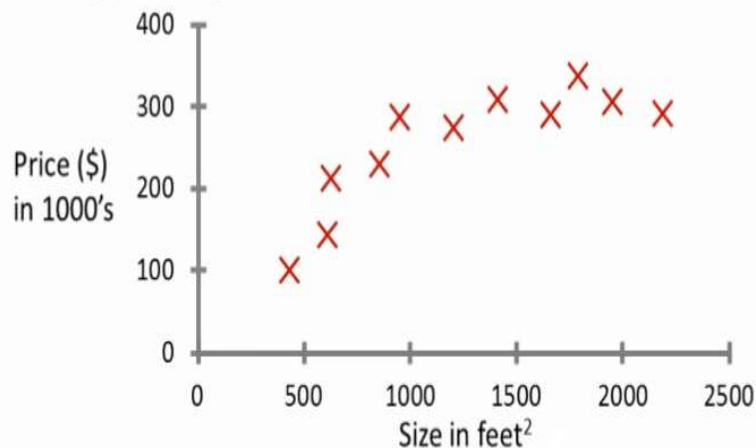
$$h(x) = y$$



Housing Price Prediction

Regression !!!

Housing price prediction.



Living area (feet ²)	Price (1000\$s)
2104	400
1600	330
2400	369
1416	232
3000	540
⋮	⋮

Learn a function h such that

$h(\text{living area in } \textit{feet}^2) \sim \text{Price in 1000\$s}$

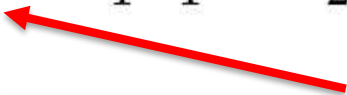
Regression vs Classification

- Regression
 - Target variable is continuous
 - Example,
 - Housing price prediction
- Classification
 - Target variable can take only small number of discrete values
 - Example,
 - Email classification (Spam vs Ham)

Linear Regression

- We assume a linear relation between X and Y .
- Given x , we approximate y with a linear function.
- Housing price prediction with two features

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

$$h_{\theta}(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$


$x_0 = 1$

- θ is called parameters/weights
 - A $d+1$ dimensional vector of weights

Living area (feet ²)	#bedrooms	Price (1000\$)
2104	3	400
1600	3	330
2400	3	369
1416	2	232
3000	4	540
⋮	⋮	⋮

Linear Regression

- Goal:
 - Given a training set, learn θ
 - Such that $h_{\theta}(x)$ is as close as possible to y
- If we had only one sample, minimize

$$J(\theta) = \frac{1}{2} (h_{\theta}(x) - y)^2$$


$$h_{\theta}(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$

Linear Regression

- Goal:
 - Given a training set, learn θ
 - Such that $h_{\theta}(x)$ is as close as possible to y

Least Squares (LS)

- For a dataset with n samples, minimize

$$J(\theta) = \frac{1}{2} \sum_{k=1}^n (h_{\theta}(x^{(k)}) - y^{(k)})^2$$

Cost Function/Loss Function

$$h_{\theta}(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$

Linear Regression

- Goal: Choose θ to minimize $J(\theta)$

$$\theta^* = \operatorname{argmin}_{\theta} J(\theta)$$

- How?

Linear Regression

$$J(\theta) = \frac{1}{2} \sum_{k=1}^n (h_{\theta}(x^{(k)}) - y^{(k)})^2$$

- Goal: Choose θ to minimize $J(\theta)$

$$\theta^* = \operatorname{argmin}_{\theta} J(\theta)$$

- Set partial derivatives to 0 to obtain minima

$$\frac{\partial J}{\partial \theta_0} = \sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) = 0$$

$$\frac{\partial J}{\partial \theta_1} = \sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) x_1^{(k)} = 0$$

$$\frac{\partial J}{\partial \theta_2} = \sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) x_2^{(k)} = 0$$

Linear Regression

- Simplifying the partial derivative equations,

$$\sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) = 0$$

$$\Rightarrow n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) x_1^{(k)} = 0$$

$$\Rightarrow \theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\sum_{k=1}^n (\theta_0 + \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} - y^{(k)}) x_2^{(k)} = 0$$

$$\Rightarrow \theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

System of Linear Equations

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

- Vectorized Form

$$\begin{pmatrix} n & \sum x_1^{(k)} & \sum x_2^{(k)} \\ \sum x_1^{(k)} & \sum (x_1^{(k)})^2 & \sum x_1^{(k)} x_2^{(k)} \\ \sum x_2^{(k)} & \sum x_1^{(k)} x_2^{(k)} & \sum (x_2^{(k)})^2 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} \sum y^{(k)} \\ \sum x_1^{(k)} y^{(k)} \\ \sum x_2^{(k)} y^{(k)} \end{pmatrix}$$

Linear Regression

- Simplifying the partial derivative equations,

$$n\theta_0 + \theta_1 \sum_k x_1^{(k)} + \theta_2 \sum_k x_2^{(k)} = \sum_k y^{(k)}$$

$$\theta_0 \sum_k x_1^{(k)} + \theta_1 \sum_k (x_1^{(k)})^2 + \theta_2 \sum_k x_1^{(k)} x_2^{(k)} = \sum_k x_1^{(k)} y^{(k)}$$

$$\theta_0 \sum_k x_2^{(k)} + \theta_1 \sum_k x_1^{(k)} x_2^{(k)} + \theta_2 \sum_k (x_2^{(k)})^2 = \sum_k x_2^{(k)} y^{(k)}$$

- Vectorized form

$$\begin{pmatrix} n & \sum x_1^{(k)} & \sum x_2^{(k)} \\ \sum x_1^{(k)} & \sum (x_1^{(k)})^2 & \sum x_1^{(k)} x_2^{(k)} \\ \sum x_2^{(k)} & \sum x_1^{(k)} x_2^{(k)} & \sum (x_2^{(k)})^2 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} \sum y^{(k)} \\ \sum x_1^{(k)} y^{(k)} \\ \sum x_2^{(k)} y^{(k)} \end{pmatrix}$$

$$(X^T X)\theta = X^T y$$

X is a $n \times (d + 1)$ matrix
Each row is a multivariate sample

y is a $n \times 1$ vector
Each row is a label of a sample

Linear Regression

- Solve the system of linear equations to get θ^*

$$(X^T X)\theta = X^T y$$

$$\theta^* = (X^T X)^{-1} X^T y$$

Normal Equation

Linear Regression

- Once trained predict housing prices with,

$$\hat{y} = (\theta^*)^T x$$

- What if $n \gg 1$??
- What if $d \gg 1$??
- What if the $(X^T X)$ is non-invertible?
- What if the relation between **X** and **y** are not linear?

Reference

- Dive into Deep Learning (d2l)
 - Sections 3.1 (up to 3.1.2), 3.4
- Deep Learning [Optional Reading]
 - Section 5.1.4